Task 3.2: Find the bug

```sql
-- Task 3.2 Find the bug
-- 3.2.1 Pick a nullable column on your event or producer table. If the dataset has no NULLs in it, insert a few deliberately and say that you did.
SELECT cancellation_reason  357 ms
FROM trips
WHERE cancellation_reason != 'Rider no show';
-- notice that the WHERE skipped the NULLS (22 items)

SELECT cancellation_reason
FROM trips
WHERE cancellation_reason = 'NULL';
-- doesn't return anything

SELECT COUNT(*) AS total_trips
FROM trips;
-- this is the total number of trips (200)

SELECT COUNT(*) AS null_count
FROM trips
WHERE cancellation_reason IS NULL;
-- Count number of Nulls (166)

SELECT COUNT(*) AS rider_no_show_count
FROM trips
WHERE cancellation_reason = 'Rider no show';
-- 12 Rider no show
-- 12 are Ruder nwo show , 22 are other reasons that are not Rider no show = 34 - there are 200 total rides so 166 did not show up
-- 12 + 22 + 166 = 200

-- 3.2.3 Write a second query counting the rows the first one silently omitted, and show that the two counts do not sum to
-- the table total.
SELECT COUNT(*) AS no_show_or_null_count
FROM trips
WHERE LOWER(cancellation_reason) = 'rider no show'
 OR cancellation_reason IS NULL;
-- 178 were Rider ho show or NULL
-- 12 Rider no show + 166 NULL = 178

--3.2.4 Repair the first query using an explicit IS NULL condition or COALESCE, so it returns what you intended
    SELECT COUNT(*) AS no_show_or_null_count
FROM trips
WHERE LOWER(cancellation_reason) != 'rider no show'
   OR cancellation_reason IS NULL;
-- 178 were Rider ho show or NULL
-- 22 are NOT Rider no show + 166 NULL = 188
-- This is what I actually would have expected to see if I ran my first query adn knew that there were 200 entries

-- If I search for the number of cancellation_reasons that are not "Rider no show" I get 22
-- If I search for the number of cancellation_reasons that are "Rider no show" I get 12
-- When I search for the total number of entries I get 200
-- Leaves one to ask where are the remaining (200 - 12 - 22) = 166 entries?
-- I can not identify where all of teh entries are
-- I should have gotten 166 + 22 rows for cancellation_reason != "Rider no show"
```

```
d
lumn on your event or producer table. If the dataset has no NULLs in it, inse
357 ms

= 'Rider no show';
ipped the NULLS (22 items)
```

Output | ex603_data.public.trips

Tx: Auto | DDL

cancellation_reason
Payment failed
Driver unavailable
Rider cancelled
Payment failed
Rider cancelled
Rider cancelled
Payment failed
Payment failed
Driver unavailable
Driver unavailable
Payment failed
Rider cancelled
Payment failed
Rider cancelled
Driver unavailable
Driver unavailable
Driver unavailable
Rider cancelled
Driver unavailable
Payment failed
Rider cancelled
Rider cancelled

```sql
SELECT COUNT(*) AS total_trips  373 ms
FROM trips;
-- this is the total number of trips (200)
```

Output | doesn't return anything

	total_trips
1	200

Tx: Auto | Playground

```sql
SELECT COUNT(*) AS null_count  352 ms
FROM trips
WHERE cancellation_reason IS NULL;
-- Count number of Nulls (166)
```

Output | this is the total number of trips (200)

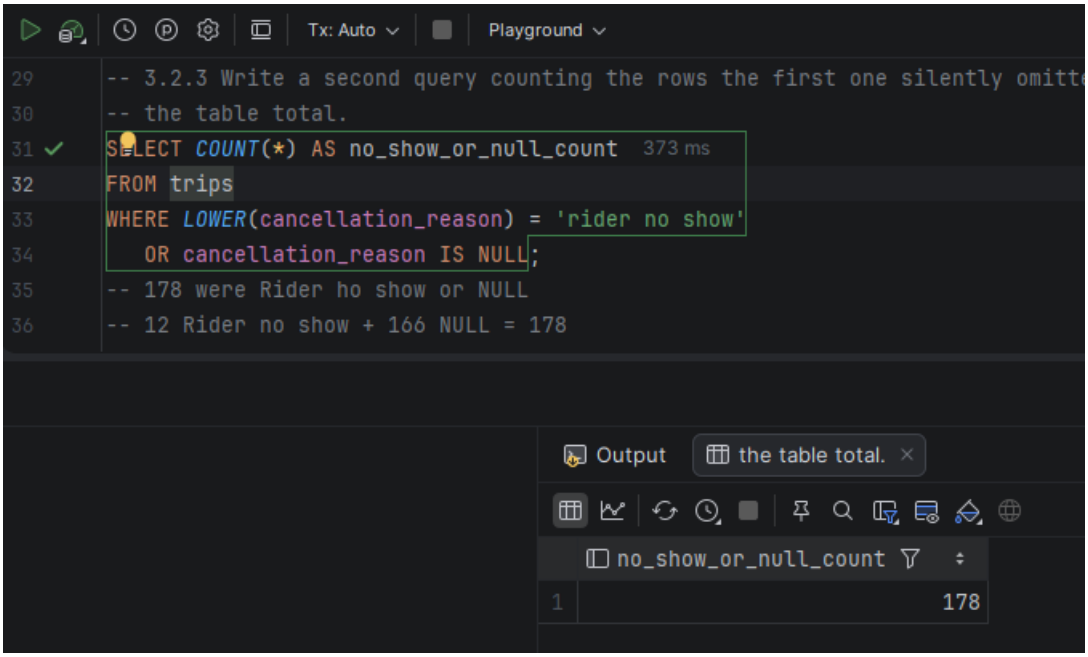
	null_count
1	166

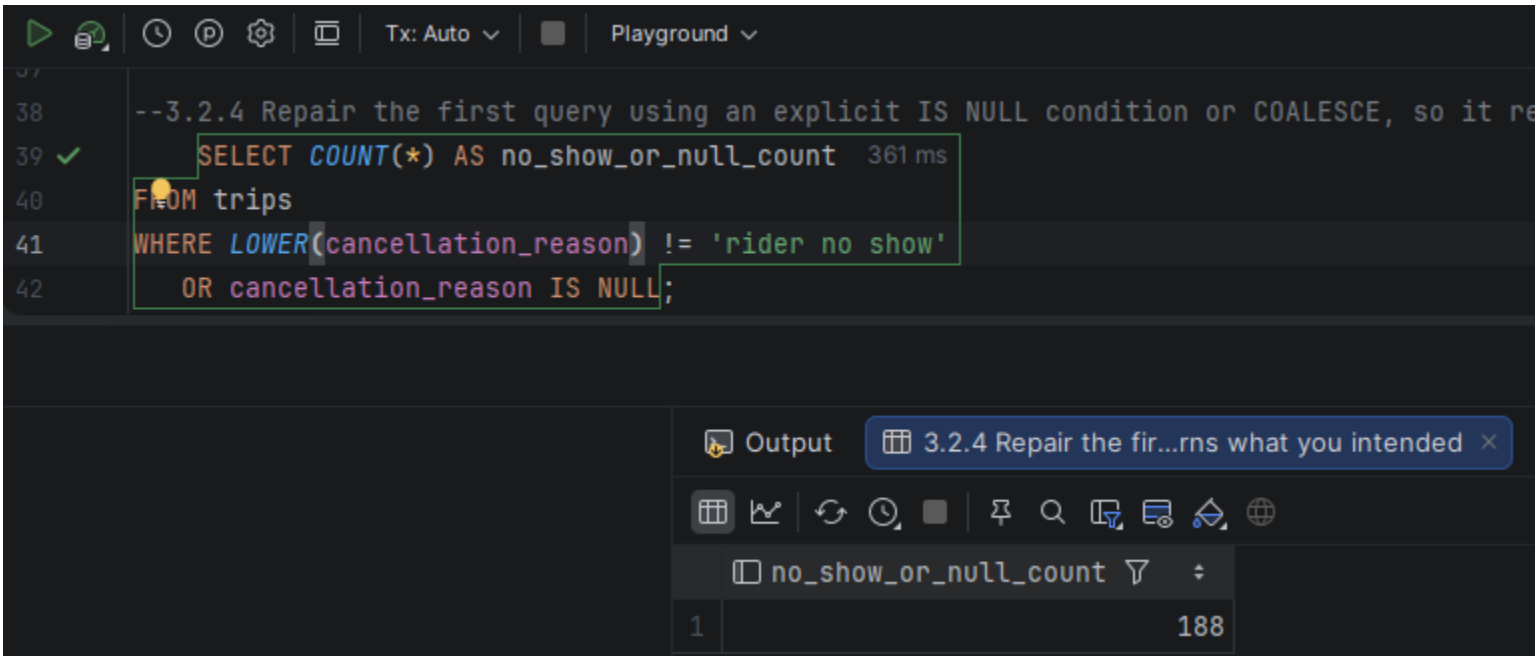
Tx: Auto | Playground

```sql
SELECT COUNT(*) AS rider_no_show_count  350 ms
FROM trips
WHERE cancellation_reason = 'Rider no show';
-- 12 Rider no show
-- 12 are Ruder nwo show , 22 are other reasons that are not Rider no show =
-- 12 + 22 + 166 = 200
```

Output | Count number of Nulls (166)

	rider_no_show_count
1	12


Tx: Auto
Playground
-- 3.2.3 Write a second query counting the rows the first one silently omitt
-- the table total.
SELECT COUNT(*) AS no_show_or_null_count 373 ms
FROM trips
WHERE LOWER(cancellation_reason) = 'rider no show'
OR cancellation_reason IS NULL;
-- 178 were Rider ho show or NULL
-- 12 Rider no show + 166 NULL = 178
Output
the table total.
no_show_or_null_count
1
178


Tx: Auto
Playground
--3.2.4 Repair the first query using an explicit IS NULL condition or COALESCE, so it re
SELECT COUNT(*) AS no_show_or_null_count 361 ms
FROM trips
WHERE LOWER(cancellation_reason) != 'rider no show'
OR cancellation_reason IS NULL;
Output
3.2.4 Repair the fir...rns what you intended
no_show_or_null_count
1
188

Part B

```
SELECT trip_id,  35 ms
       fare_amount / NULLIF(distance_km, 0) AS fare_per_km
FROM trips
WHERE fare_per_km > 3;
```

```
[42703] ERROR: column "fare_per_km" does not exist
  Position: 96
```

Output | 3.2.4 Repair the fir...rns what you intended

	no_show_or_null_count
1	188

```
SELECT trip_id,
       fare_amount / NULLIF(distance_km, 0) AS fare_per_km
FROM trips
WHERE fare_per_km > 3;

-- Rewrite 1: Repeat the expression in WHERE
SELECT trip_id,
       fare_amount,
       distance_km,
       ROUND(fare_amount / NULLIF(distance_km, 0), 2) AS fare_per_km
FROM trips
WHERE fare_amount / NULLIF(distance_km, 0) > 3
ORDER BY trip_id;

-- Rewrite #2: A CTE as the alias becomes a real column

WITH trip_rates AS (  400 ms
    SELECT trip_id,
           fare_amount,
           distance_km,
           ROUND(fare_amount / NULLIF(distance_km, 0), 2) AS fare_per_km
    FROM trips
)
SELECT trip_id, fare_amount, distance_km, fare_per_km
FROM trip_rates
WHERE fare_per_km > 3
ORDER BY trip_id;
```

Output | Rewrite #2: A CTE as...becomes a real column

	trip_id	fare_amount	distance_km	fare_per_km
1	123	4.70	1.50	3.13